Conditional probability

1 $\frac{1}{3}$

2

a 0.23

b 0.73

c $\frac{27}{50}$

d $\frac{23}{56}$

e $\frac{17}{50}$

3

a $x = 0$

b $x = 0.2, 0.3$

c $x = 0.3$

4

a

	2	4	5
3	5	7	8
7	9	11	12

b $\frac{2}{5}$

c 1

5

a

	1	2	4	6
1	1	2	4	6
2	2	4	8	12
3	3	6	12	18
4	4	8	16	24
5	5	10	20	30
6	6	12	24	36

b $\frac{1}{2}$

c $\frac{3}{10}$

d $\frac{1}{2}$

Dependent and independent events

6

a Independent

b Dependent

c Dependent

d Dependent

7

a Yes

b No

8 0.2

9

a

i 0.3

ii 0.2

iii 0.1

iv Non-mutually exclusive

b

i 0.2

ii 0.1

iii 0

iv Mutually exclusive

c

i 0.6

ii 0.1

iii 0.2

iv Non-mutually exclusive

d

i 0.1

ii 0.7

iii 0

iv Mutually exclusive

10

a

i 0.42

ii 0.12

iii 0.18

iv 0.18

v Yes

b

i 0.15

ii 0.05

iii 0.05

iv 0.02

v No

11

a 0.8

b 0.05

c $P(B) = x + 0.05$ or $P(B) = \dfrac{x}{0.8}$

d $x = 0.2$

e 0.25

12

a 0.7

b Yes

c 0.5

13 $\dfrac{8}{33}$

14

a $\frac{32}{663}$

b $\frac{4}{39}$

15

a

i $\frac{33}{16\,660}$

ii $\frac{1}{649\,740}$

b $2\,598\,960$

16

a No

b $\frac{1}{6}$

17

a $\frac{1}{45}$

b $\frac{1}{44}$

c No

d Dependent

18

a $\frac{1}{51}$

b $\frac{1}{50}$

c Dependent

19

a $\frac{5}{18}$

b $\frac{1}{3}$

20

a $\frac{1}{64}$

b $\frac{11}{850}$

21

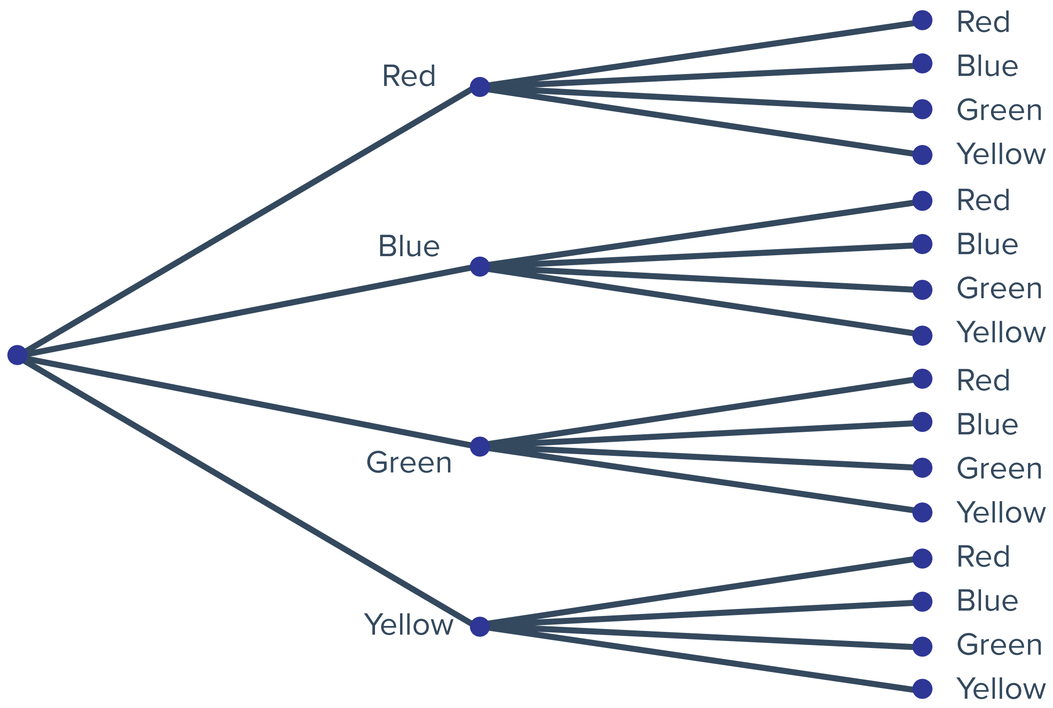
a $\frac{1}{3}$

b $\frac{1}{12}$

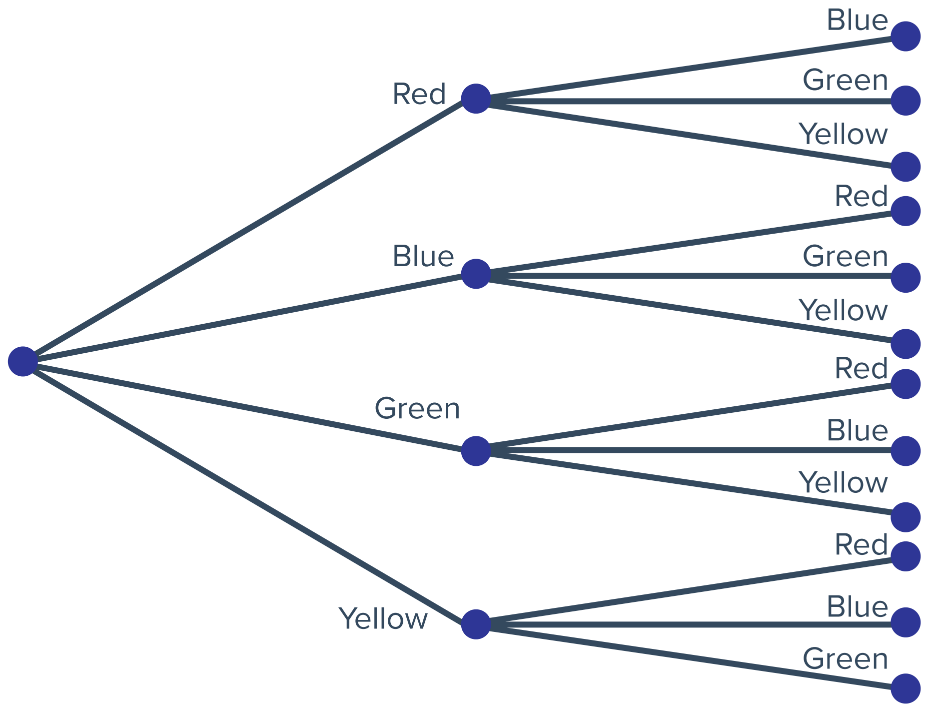
Tree diagrams and probability trees

22

a

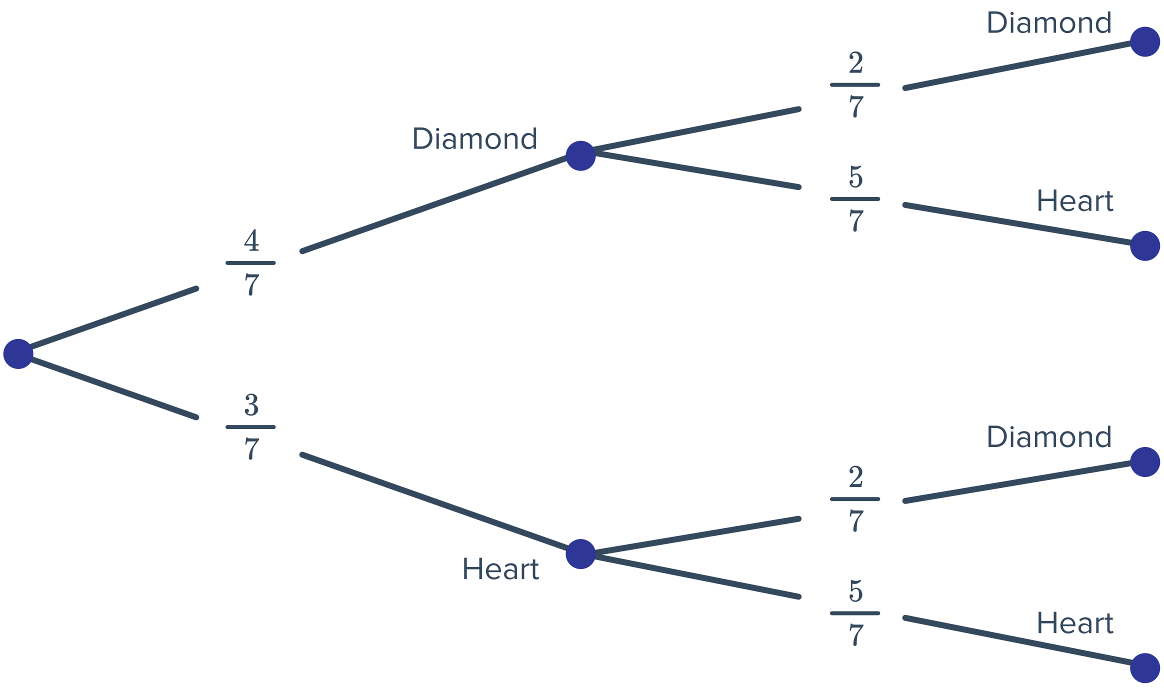


b



23

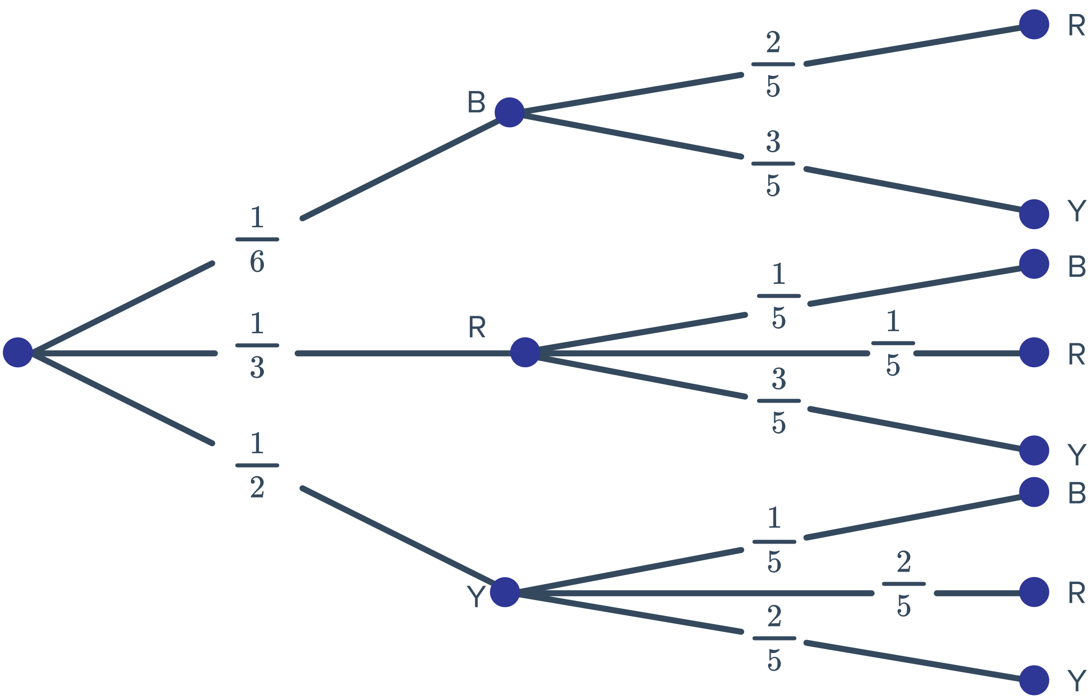
a



b $\frac{15}{49}$

24

a

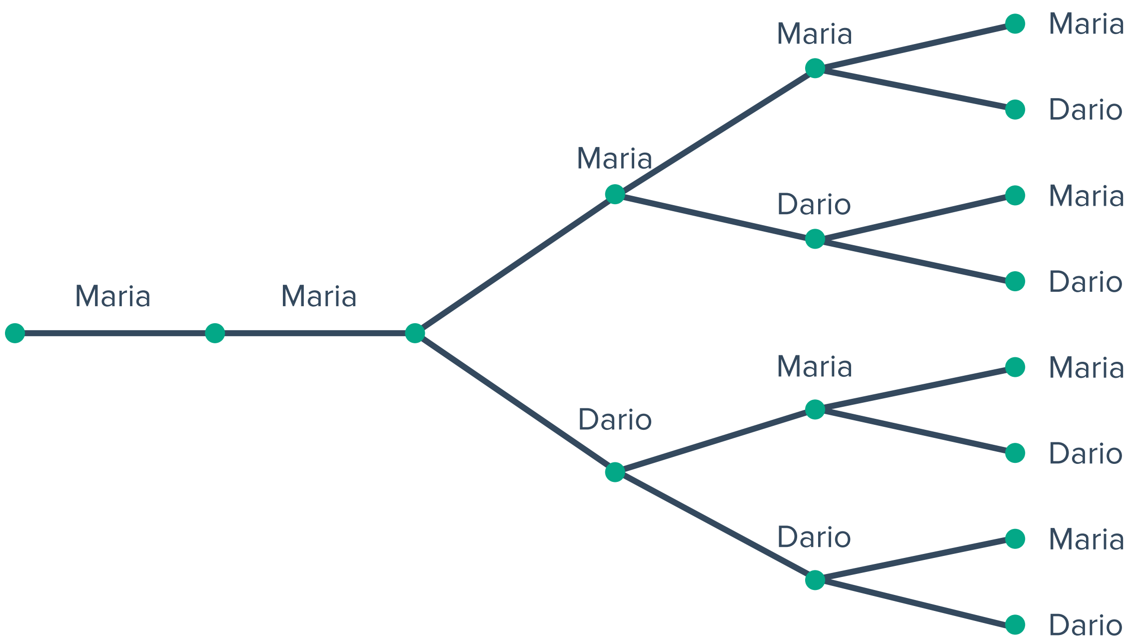


b

i $\frac{1}{10}$ **ii** $\frac{1}{15}$ **iii** $\frac{1}{15}$ **iv** $\frac{1}{5}$

v 0 **vi** $\frac{1}{5}$ **vii** $\frac{2}{5}$

25 **a**

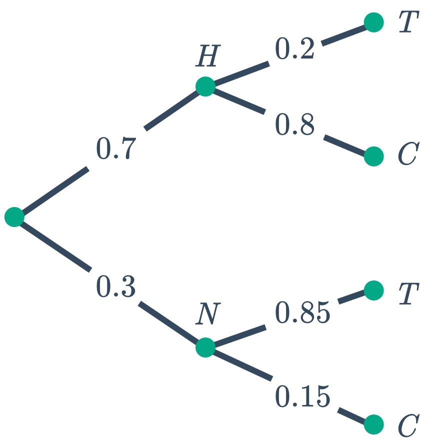


b $\frac{2}{4}$

c $\frac{1}{4}$

26

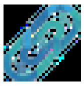
a

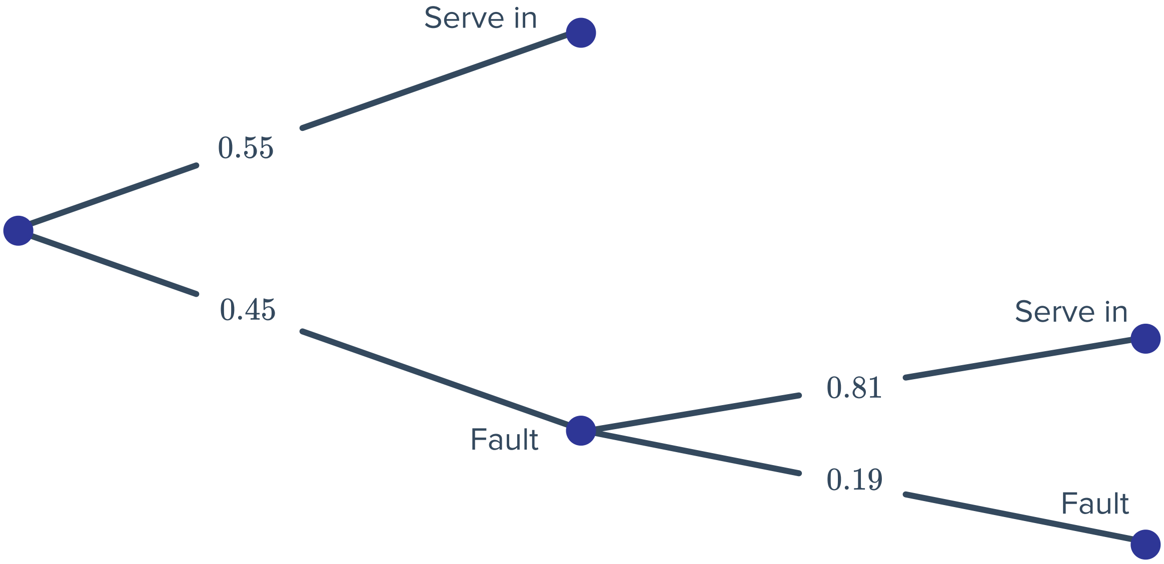


b 0.35

27

a



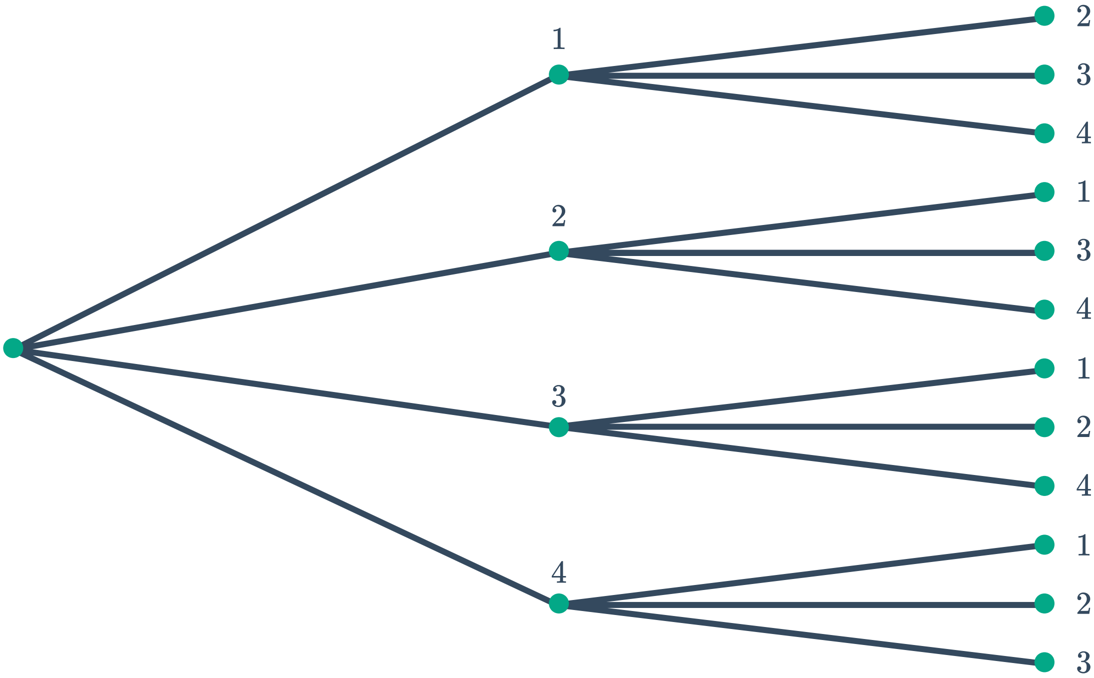


b 0.45

c 0.0855

28

a



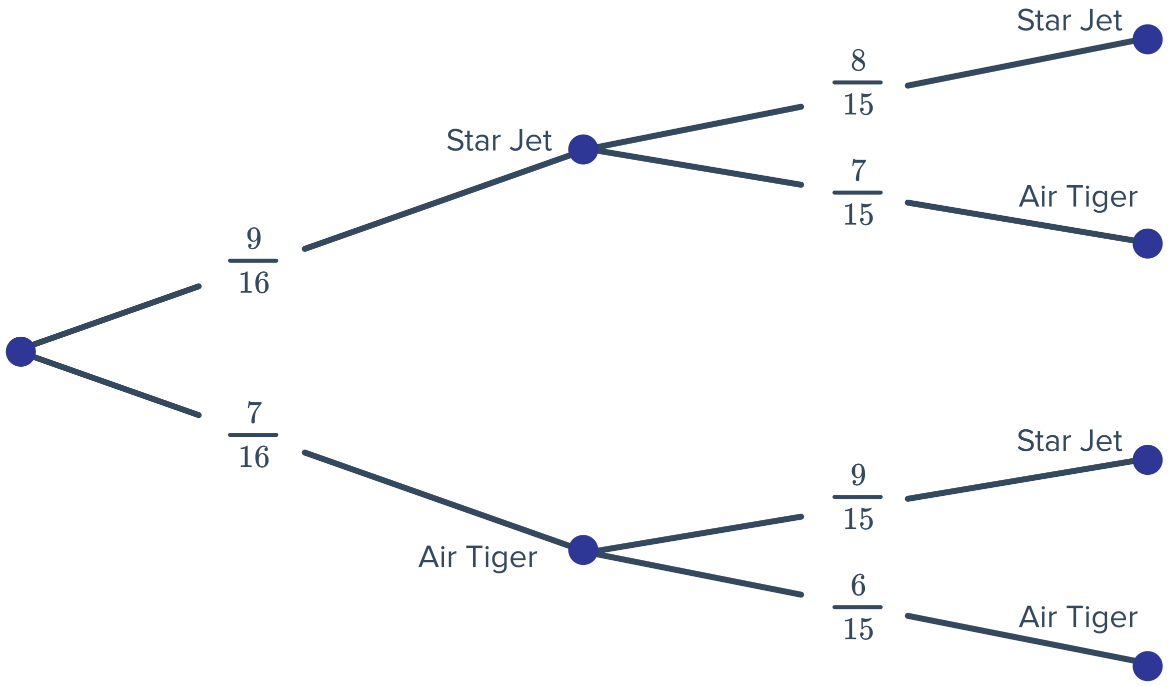
b

i $\frac{1}{2}$ ii 0 iii $\frac{1}{2}$ iv $\frac{1}{3}$

v 0 vi $\frac{1}{4}$ vii $\frac{1}{4}$

29

a



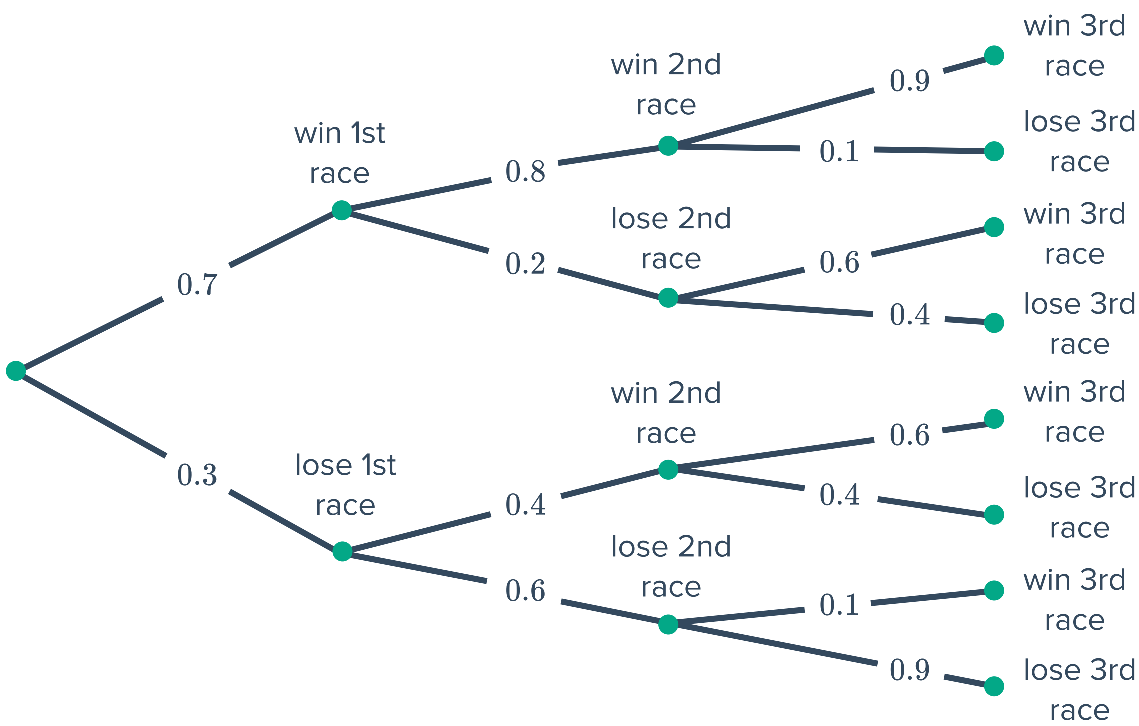
b

i $\frac{19}{40}$

ii $\frac{21}{40}$

30

a



b 0.504

c 0.678

Applications

31

a $\frac{1}{52}$

b $\frac{1}{52}$

c $\frac{1}{52}$

d $\frac{1}{52}$

e The probability is getting higher.

32

a $\frac{1}{55}$

b $\frac{1}{55}$

c Independent

33

a $\frac{1}{31}$

b Dependent

34

a $\frac{1}{3}$

b $\frac{1}{11}$

c $\frac{41}{55}$

35

a $\frac{1}{6}$

b $\frac{25}{174}$

c $\frac{6325}{71\,253}$

d The probability is getting higher.

e 25

36

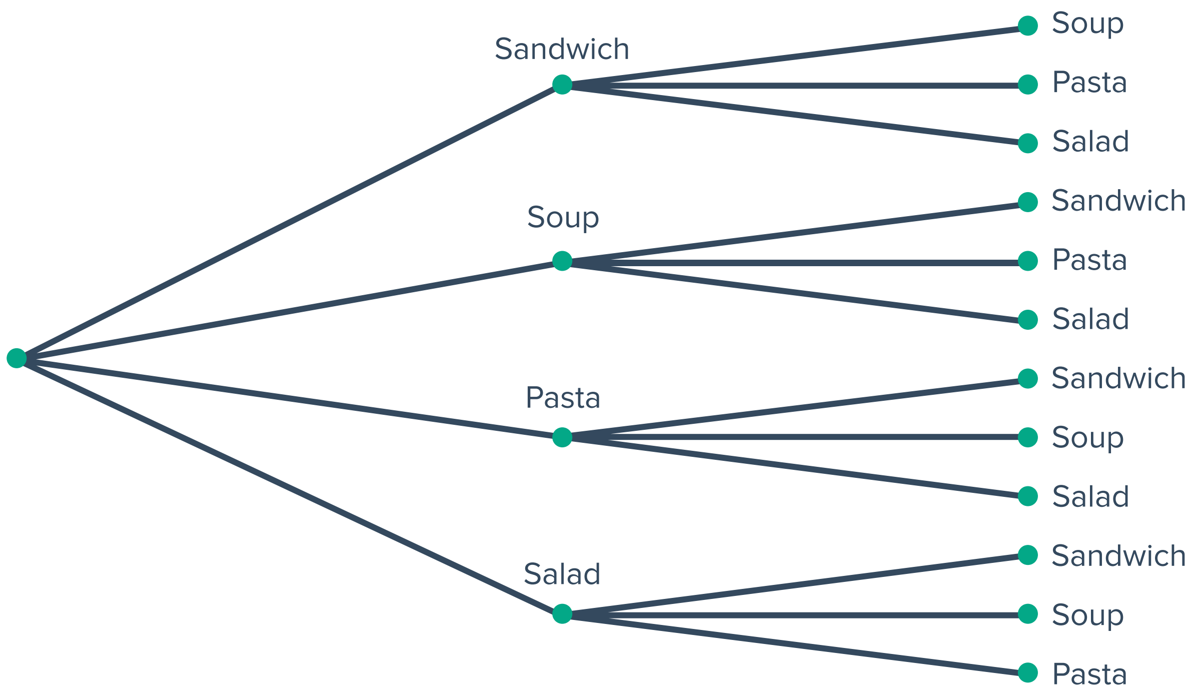
a $746, 764, 476, 467, 674, 647$

b $\frac{1}{2}$

c $\frac{1}{2}$

37

a



b

i $\frac{1}{12}$ **ii** $\frac{1}{6}$ **iii** $\frac{1}{2}$ **iv** $\frac{1}{2}$

v $\frac{1}{12}$

38

a $2, 4, 6, 8, 10, 12, 14, 16, 18, 20$

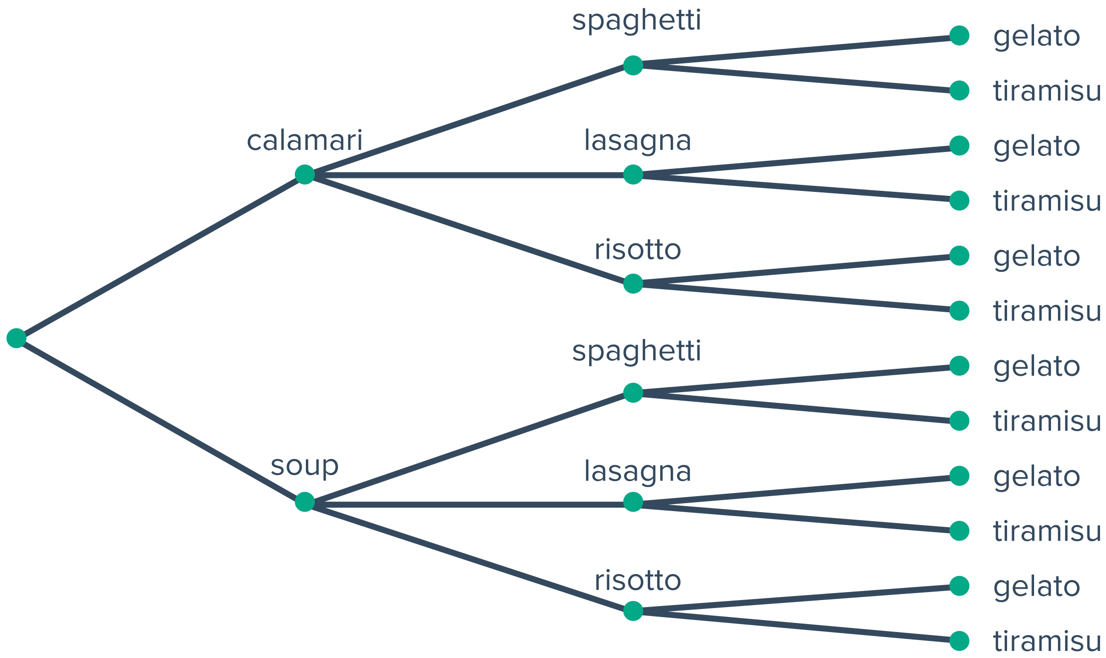
b $1, 2, 3, 4, 6, 8, 12, 16$

c 6

d $\frac{3}{5}$

39

a



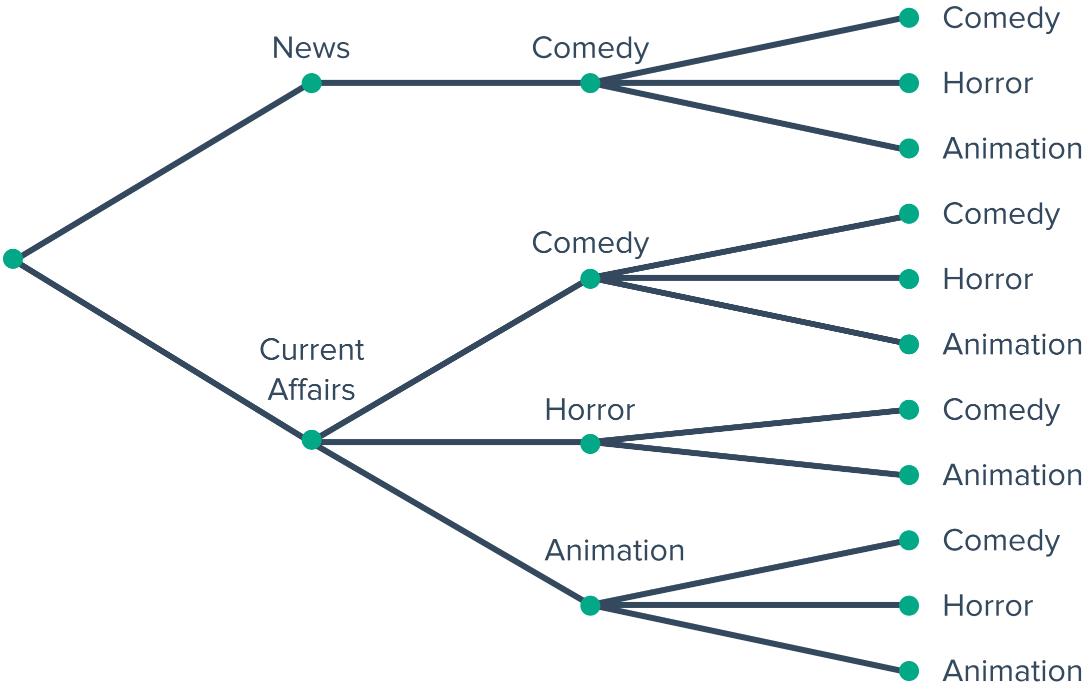
b $\frac{1}{3}$

c $\frac{1}{6}$

d $\frac{3}{4}$

40

a



b $\frac{1}{3}$

41

a

	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	26	41	3	70
Liberal (last election)	15	74	1	90
Greens (last election)	9	25	6	40
Total	50	140	10	200

b $\frac{1}{6}$

c $\frac{2}{5}$

d $\frac{23}{32}$

42

a $\frac{1}{9}$ b $\frac{5}{9}$ c $\frac{1}{3}$

43

a

i 260 ii 237 iii 111 iv 208

b $\frac{208}{319}$

44

a $\frac{16}{69}$ b 10 c $\frac{5}{13}$ d $\frac{2}{3}$

45

a 25 b $\frac{17}{87}$ c $\frac{25}{42}$ d $\frac{17}{27}$

46

a 3 b $\frac{47}{50}$ c $\frac{3}{10}$

47

a

	Read the book	Didn't read the book	Total
Watched the movie	39	58	97
Didn't watch the movie	65	4	69
Total	104	62	166

b

i $\frac{65}{166}$ **ii** $\frac{39}{97}$ **iii** $\frac{29}{31}$

48

a

	Believe in Santa	Do not believe in Santa	Total
Believe in Tooth Fairy	31	10	41
Do not believe in Tooth Fairy	49	10	59
Total	80	20	100

b 38.75%

c $\frac{31}{90}$

d $\frac{10}{41}$

49

a

i $x = 426$ **ii** $y = 398$ **iii** $z = 154$

b $\frac{199}{489}$

c $\frac{77}{276}$

50

a

	Paperback	Not Paperback	Total
E-Reader	60	84	144
Not E-Reader	20	36	56
Total	80	120	200

b $\frac{21}{50}$

c 42%

51

a

i $x = 40$ **ii** $y = 400$ **iii** $z = 200$

b 360

c $\dfrac{10}{19}$